

SACHIN KUMAR

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EXECUTIVE SUMMARY

AI Automation Engineer at TCS with experience building AI/ML systems for incident management, data retrieval, and infrastructure operations. Skilled in Python, SQL, LLMs, RAG, multi-agent systems, and machine learning, with hands-on experience in LangGraph, LangChain, MCP, GCP, and Azure. Built production solutions that reduced MTTR from 4 hours to 20 minutes and server processing time by 99%. Experienced in developing, deploying, and evaluating AI systems with measurable improvements in operational efficiency.

EDUCATION

Vellore Institute of Technology, Vellore, India

Master of Science in Business Statistics

DG Vaishnav, Chennai, India

Bachelor of Commerce - General

Sep 2022 - May 2024

GPA: 9.02/10

Jun 2019 - May 2022

GPA: 7.45/10

EXPERIENCE

AI Engineer

Jan 2025 - Present

Tata Consultancy Services (TCS)

Chennai, India

- Designed a **Python** and **SQL** auto-resolution engine that retrieves **ServiceNow** incidents and executes automation scripts, reducing MTTR from **4 hours to 20 minutes** without human involvement.
- Developed ITSM automation tool leveraging **LLM** technology to streamline incident management.
- Orchestrated a multi-agent workflow with **LangGraph** and an **NL-to-SQL** engine in **Python**, compressing ad hoc data retrieval from hours to seconds.
- Architected a hybrid **retrieval-augmented generation** pipeline using **FAISS**, **BM25**, and **cross-encoder reranking**; created statistical evaluation frameworks to benchmark retrieval accuracy against a curated ground-truth dataset.
- Developed and deployed a custom **MCP** server for secure, natural-language-driven **SQL** **CRUD** operations across relational databases, resulting in improved efficiency and managing 36% more database transactions.
- Led deployment of a **time-series forecasting** system using **MSTL**, **Linear Regression**, and **XGBoost** in **Python** to anticipate CPU and memory issues two hours in advance and prevent outages.
- Reduced server processing time by 99%, from **4 minutes to under 3 seconds**, through iterative experimentation and algorithmic refinement, enabling infrastructure scaling.
- Architected a clustering-based **correlation system** that maps system dependencies and groups duplicate alerts in real time, reducing incident response time by 45%.

PROJECTS

Multi-Agent Debate & Consensus System | *Python, SQL, FastAPI, GCP Cloud Run, Pub/Sub, BigQuery, Docker*

- Built and deployed a **Gemini 2.0** Flash multi-agent system on **GCP** using **Python**; benchmarked response times and optimized asynchronous **Pub/Sub** event streams, reducing end-to-end latency by 38%.
- Designed four specialized AI agents as independent **Cloud Run** microservices, communicating and coordinating through **Pub/Sub** and persisting distributed state in **BigQuery** with **SQL** for downstream analysis.

HR Assistant MCP Agent | *Python, MCP, Claude Desktop, LangChain, Git*

- Digitized new-employee onboarding through a custom **Python**-based **MCP** server integrated with **Git** version control, eliminating manual portal interactions.
- Integrated **HRMS** workflows for leave management, ticketing, and meeting scheduling as **LLM-callable** **Python** functions, enabling natural-language process automation.

AI Marketing Content Crew | *Python, CrewAI, Prompt-Engineering, YAML, Git*

- Built a multi-agent AI system using **Python**, **CrewAI**, and the **Google Gemini LLM** to automate the marketing content pipeline from initial web research to final drafting.
- Engineered collaborative agents with integrated web-scraping tools and persistent memory to autonomously generate structured, audience-tailored content with minimal manual intervention.

SKILLS

Languages: Python, SQL (PostgreSQL, MSSQL, MySQL)

AI / LLM Engineering: LLMs, RAG, Agentic & Multi-Agent Systems, Prompt Engineering, LLM Fine-Tuning, OpenAI, Semantic Search, Vector Databases (FAISS, ChromaDB, Pinecone), Model Context Protocol (MCP)

ML & Data Science: Machine Learning, Deep Learning, NLP, Time-Series Forecasting, Statistical Modelling, Model Evaluation

Cloud & MLOps: GCP (Vertex AI, BigQuery, Cloud Run), Azure (Azure OpenAI, AI Foundry, Functions), Docker, Git, CI/CD, Databricks

Frameworks & Libraries: LangChain, LlamaIndex, LangGraph, Google ADK, Hugging Face, PyTorch, TensorFlow, Scikit-learn, REST API

CERTIFICATIONS

Cloud & AI: Azure AI Engineer Associate (AI-102) • Azure AI Fundamentals (AI-900) • Google Cloud Associate Cloud Engineer

GenAI & Data: Claude Certified Developer – Foundations • Claude Certified Associate – Foundations • IBM Data Science Professional • Generative AI for Developers